

Use the Right Tools to Debug Java Applications

Debugging code is an essential part of the program development cycle, and debugging Java applications has its own special problems. In this article, the author analyses the causes of problems in Java code, and suggests ways and means to debug them.



Java programs are always complex to debug and analyse due to their versatility in supporting a lot of features, platforms, etc. When writing Java programs, one of the primary concerns the developer has to address is the provision of a debug facility for future use, to solve any problem that may occur in a test environment or even in a real-time client environment.

The debugging facility would depend on the types of problems that might arise during Java program execution. These problems would depend upon:

1. Features used (like device interface, external connectivity like file handling, database usage)
2. Concept implemented/linked (like RMI, JDBC, threads, communication with other Java programs)
3. GUI facility

4. User interaction with a utility, script, GUI, etc
5. Compatibility (like the OS handled, JDK version supported, etc)

Classification of Java problems

Based on these factors and other minor categories, we can classify problems in Java applications as follows:

1. Memory related problems
2. Synchronisation/thread related issues
3. Communication related problems (backend, interface and GUI)
4. Problems related to the database, connectivity and file handling
5. Issues linked to the boundary condition
6. Native/JNI related issues